

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

INJECTION MOLD DESIGN
MODEL QUESTION PAPER – SET-I

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

Module Outcome Cognitive level

1	Cast iron normally contains _____ % carbon?	M1.01	R
2	The letters EN in steel standard stands for _____ ?	M1.02	U
3	Which machine is known as mother of machine tools?	M1.03	U
4	Core is normally positioned in _____ platen of injection moulds?	M2.01	U
5	Name of the alloy used to deposit over copper substrate in electro-deposition method?	M2.02	U
6	Small blind well provided opposite to sprue in an injection mould is known as _____?	M2.03	U
7	For complex parting surface moulds _____ runner cross-section is preferred?	M3.01	U
8	Jetting is a filling problem associated with _____ gate?	M3.02	U
9	To prevent coolant leakage through two adjacent plates in a mould half _____ is used?	M4.04	U

PART B

II. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

Module Outcome Cognitive level

1	List the factors determining the selection of mould material for a particular mould?	M1.02	U
2	What are the turning operations that can be performed in a lathe?	M1.03	U
3	Illustrate the operations that can be carried out using a planer and shaper?	M1.03	U
4	Using a sketch identify the core, cavity and impression of an injection mould?	M2.01	U
5	Write a short note on bolsters and their requirements?	M2.02	U
6	Outline the methods used to attach mould halves to machine platens?	M2.03	U
7	Using a figure describe the feed system of injection mould?	M3.01	U
8	Describe different runner cross-sections used in injection moulds?	M3.02	U
9	Show different U-shaped cooling circuit constructions used in injection moulds?	M4.03	U
10	Briefly explain the use of O-rings in injection mould?	M4.04	U

PART C

III. Answer all questions from the following

(6x 7 = 42 Marks)

Module Outcome Cognitive level

1	Using a line diagram describe the functions of parts of a lathe?	M1.03	U
OR			

2	Describe the working of a planer using a neat sketch?	M1.03	U
3	Using a block diagram outline the working of a tracer controlled milling machine?	M1.03	U
OR			
4	Illustrate the Shaw investment casting method for making core inserts?	M2.02	U
5	Show the steps involved in cold hobbing process for making cavity inserts?	M2.02	U
OR			
6	Discuss the type of register rings used in injection moulds?	M2.03	U
7	Account the factors considered while designing the size of a runner?	M3.02	U
OR			
8	Discuss on the designing features of rectangular edge gate?	M3.02	U
9	Describe spring loaded ejector plate assembly return mechanism?	M3.04	U
OR			
10	Briefly explain parting surfaces of injection moulds?	M4.01	U
11	Outline different methods used for cooling integer type core plate?	M4..03	U
OR			
12	Explain cooling circuits recommended for sprue bush?	M4.03	U

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

INJECTION MOULD DESIGN
MODEL QUESTION PAPER – SET-2

Time: 3 hours

Maximum Marks: 75

PART A

IV. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	Iron containing _____ % of carbon is known as steel?	M1.01	R
2	In lathe work, when the tool is fed in an angle to the rotation of job, it will produce a _____ surface?	M1.03	U
3	Suggest a machine tool suitable for producing contour shapes at better accuracy and surface finish?	M1.03	U
4	Part of the mould which imparts the shape of the product is called _____?	M2.01	R
5	Nozzle of an injection machine is connected to the injection mould cavity through _____?	M2.03	U
6	Sprue-Runner-Gate manifold of an injection mould is known as _____?	M3.01	R
7	Part of the mould which provide a space into which the ejector plate assembly can be fitted and operated is called _____?	M3.03	U
8	The meeting point of two flow paths around a core in an impression is known as _____?	M3.02	U
9	For cooling Integer type cavity plate from all sides _____ cooling circuit is more suitable?	M4.03	U

PART B

V. Answer any Eight questions from the following

(8 x 3= 24 Marks)

Module Outcome Cognitive level

1	Using a sketch explain the working principle of a lathe?	M1.03	U
2	Distinguish between shaper and planer?	M1.03	U
3	Write a short note on methods of grinding?	M1.03	U
4	List various steps involved in bench fitting of an injection mould?	M2.04	U
5	Write a short note on types of sprue bush used in injection moulding?	M2.03	U
6	Justify the reason for normally keeping the core half of injection moulds in moving half?	M2.01	U
7	Sketch different shapes of runner cross-sections and conclude selection of a particular shape?	M3.02	U
8	Write down the empirical relation for gate depth and material constants for general purpose thermoplastics?	M3.02	U
9	Justify the need of venting of injection moulds?	M4.02	U
10	Illustrate the methods used for cooling shallow core inserts?	M4.03	U

PART C

VI. Answer all questions from the following (6x 7 = 42 Marks)

Module Outcome Cognitive level

1	List different machining operations possible using a lathe and briefly explain any four operations using sketches	M1.03	U
OR			
2	Sketch a cylindrical grinding machine and describe its operation?	M1.03	U
3	Using a block diagram describe the parts of a shaping machine?	M1.03	U
OR			
4	Sketch a simple injection mould and label its parts?	M2.03	U
5	List advantages and disadvantages of integer and insert type injection moulds?	M2.02	U
OR			
6	Illustrate electro-deposition process of cavity insert making?	M2.02	U
7	Illustrate different runner layouts and describe runner balancing?	M3.02	U

OR			
8	Briefly explain types of gates using simple sketches?	M3.02	U
9	Using a sketch describe different elements of an ejector plate assembly	M3.03	U
OR			
10	Discuss the requirement of balancing of mould surfaces?	M4.02	U
11	Outline different methods used for cooling cavity inserts?	M4.03	U
OR			
12	Explain cooling circuits recommended for valve type ejectors?	M4.03	U